

HOWARD UNIVERSITY
DEPARTMENT OF ECONOMICS

CODE NUMBER _____

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DATE _____

COMPREHENSIVE EXAMINATION: Fall 2024 Microeconomics Theory PhD

EXAMINERS:

- 1. Dr. Omari H. Swinton, Chairperson**
- 2. Dr. Alex Henke**
- 3. Dr. Tingting Xiong**

- 1. The examination is scheduled between the hours: 9:30 a.m.-1.00 pm**

ALL STUDENTS ARE TO BE SEATED BY 9:15 a.m.

- 2. YOU ARE REQUIRED TO ANSWER ONLY FIVE (5) QUESTIONS.**

Any additional questions answered over the required number from each category will NOT receive credit.

- 3. Correct answers to questions NOT asked will receive NO credit.**

- 4. Be sure to write the Code Number assigned to you in the TOP LEFT HAND CORNER OF THIS SHEET AND ON EACH ANSWER SHEET. DO NOT WRITE YOUR NAME ON ANY SHEET OF THE EXAMINATION.**

- 5. Begin each question on a new page. Number each page used in sequence. Write only on one side of the paper.**

- 6. Write clearly and illustrate your answers with graphs whenever and wherever possible.**

- 7. USE ONLY BLACK INK PENS.**

- 8. At the end of the examination, please indicate the total number of pages being submitted in the space provided in the TOP RIGHT HAND CORNER of this sheet.**

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- 1. Bring your pens, pencils, calculators and rulers.**

2. No briefcases, book bags or sacks, no handbags larger than 10 x 6 of any form are to be brought into the examination room.
3. No books, notes or other study material are to be brought into the examination room.
4. During the Examination there is to be no communication between or amongst students for any purpose. All questions must be directed to and channeled through the faculty member conducting the examination.
5. Only the scrap paper provided by the proctor is to be used for the examination. Scrap paper should bear the code number assigned to each student, and be handed over to the proctor along with the examination.
6. Students are not expected to leave the examination room before completing their examination and turning it in to the proctor.
7. NO FOOD OR SMOKING is permitted in the examination room.
8. It is the student's responsibility to remove any coffee or water containers taken into the examination room.
9. NO CELL PHONES ARE ALLOWED.
10. EXAMINATION RESULTS WILL ONLY BE GIVEN TO STUDENTS WHO ARE REGISTERED.

CODE NUMBER_____

STUDENTS: PLEASE CIRCLE ONLY THE QUESTIONS ANSWERED AND PROVIDE THE PAGE NUMBERS.

QUESTIONS	PAGE NUMBERS
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1.	
2.	
3.	
4.	
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6.	
7.	
8.	

1. Duality

Let $u(x,y)=(x+2)y$. Find the following

- the Marshallian demand functions for x and y.
- the indirect utility function
- the compensated (Hicksian) demand functions for x and y
- the expenditure function
- Use your answers to a and d to derive the Hicksian demand functions for x and y. Compare these to your answers to c. Explain what you are doing.
- Use your answers to b and c to derive the Marshallian demand functions. Compare these to your answers for a. Explain what you are doing.
- Write the Slutsky equation for this problem.

2. A recent Duke graduate has opened a factory for production of textbooks in Durham. The production function is as follows

$$f_d(K;L) = K^{1/3} L^{1/3}$$

Currently, she uses $K = 2000$ and $L = 500$ and the rental rates of capital and labor are $w=r=\$1$. Now she is considering expanding her business and opening another factory in Chapel Hill. Using her microeconomic skills, she estimates that the cost function for the factory in Chapel Hill will be

$$C_h(q) = q^{3/2}$$

The firm is facing a competitive market and the price at which it can sell its output is $p = \$30$.

- Suppose that in the short-run, both inputs in the Durham factory are fixed. What would be the profit maximizing output in the Chapel Hill factory? What would be the total output Q_{SR} of the firm? Calculate the profit π_{SR} that the firm will make in the short run.
- In the long run, both capital and labor are adjustable. Find the long-run cost function $C_d(q)$ for the factory in Durham.
- Suppose that the firm wants to produce Q units in total from the two factories in the long-run. How should production be allocated between the two factories in order to minimize the cost of production? Find the total cost $C_{LR}(Q)$ associated with this allocation.
- How much output Q_{LR} would the firm choose to produce in the long-run given $p = \$30$? What would be the firm's profit π_{LR} ? How does this compare to the short-run profits you found in part b?
- Describe the changes that took place in order to reach the long run equilibrium.

- Chick-Fila (A) and the McDonald's(M) are considering introducing a new drink on Howard's campus that was proven to increase substantially students' performance on exams. The two would be competing Cournot.

The demand for the drink depends on how successful A and M are in advertising in the Hilltop.

- if both of them advertise, then the demand is $P(q_A, q_M) = 6 - q_A - q_M$.
- if both of them choose not to advertise, then the drink will not be so popular and the demand will be $P(q_A, q_M) = 3 - q_A - q_M$
- if one them chooses to advertise and the other one does not, then they face the following demand functions:

$$p_i = 6 - q_i - q_j$$

$$p_j = 3 - q_i - q_j \text{ if } i \text{ advertises and } j \text{ does not.}$$

The Hilltop charges $a = \frac{1}{2}$ for advertisement.

The two competitors choose simultaneously whether to advertise and how much to produce.

- If it is known that both advertise in equilibrium, what is the equilibrium Cournot quantity (q_A^*, q_M^*) , price (p^*) and profits (π_A^*, π_M^*) for A and M?
- If it is known that both do not advertise in equilibrium, what is the equilibrium Cournot quantity (q_A^{**}, q_M^{**}) , price (p^{**}) and profits (π_A^{**}, π_M^{**}) for A and M? Are the two competitors better off under advertisement or no advertisement?
- Suppose now that the managers of A and M meet and agree not to advertise and produce (q_A^{**}, q_M^{**}) , A's manager knows for a fact that the competitor is going to stick to the agreement since he is well known for his honesty. Would A choose to

- deviate from the agreement and advertise if he only cares about profits? If yes, how much will A produce? What would be the profits for A and M?
- d. Given your answer to part (c), do you think that not advertising could ever be sustained as an equilibrium strategy if the two could not sign a binding contract?
4. Suppose the production possibility frontier for an economy that produces one public good (y) and one private good (x) is given by

$$x^2 + 100y^2 = 5,000$$

This economy is populated by 100 identical individuals, each with a utility function of the form

$$\text{Utility} = (x_i y)^{1/2}$$

where x_i is the individual's share of private good production ($=x/100$). Notice that the public good is nonexclusive and that everyone benefits equally from its level of production.

- If the market for x and y were perfectly competitive, what levels of those goods would be produced? What would the typical individual's utility be in this situation?
 - What are the optimal production levels for x and y? What would the typical individual's utility level be? How should consumption of good x be taxed to achieve this result? (Hint: The numbers in this problem do not come out evenly, and some approximations should suffice.)
5. Solve the following game for its Nash equilibrium/equilibria.

		Player 2				
		Crime	Action	Thriller	Comedy	Documentary
Player 1	Crime	10, 10	4, 3	4, 2	4, 1	4, -10
	Action	3, 4	9, 9	3, 2	3, 1	3, -10
	Thriller	2, 4	2, 3	8, 8	2, 1	2, -10
	Comedy	1, 4	1, 3	1, 2	5, 5	1, -10
	Documentary	-10, 4	-10, 3	-10, 2	-10, 1	-10, -10

6. Consider a two-stage version of the full-information symmetric all-pay auction. In the first stage $n \geq 2$ identical risk-neutral potential participants who simultaneously decide whether to pay an entry fee $\varepsilon \in (0, v)$ in order to participate in the auction, where v is the common value of the indivisible object being sold. In the second stage, each participant observes the number, m , of participants and then submits his $b \geq 0$. Non-participants take no action in the second stage and simply receive a payoff of zero.
- Derive the symmetric subgame perfect Nash equilibrium of this game.
 - What is the *ex ante* expected equilibrium payoff to each potential participant?

- c. Suppose that the seller of the object is risk neutral, but that she places value zero on retaining the object. Calculate her *ex ante* expected payoff in the subgame-perfect equilibrium. If she could choose the value of ε , what value would she set?
7. Consider a symmetric independent private values setting with two buyers. The seller owns an indivisible object which she is commonly known to value at zero. Suppose that $v_i \in \{1,2,3\}$ for $i = 1,2$, and that each realization is equally likely (i.e., occurs with the probability $1/3$). Suppose also that $S_i = \{\text{NO}, 1, 2, 3\}$ for $i = 1,2$. That is a bidder may choose not to participate or he may bid one of his three possible valuations. Finally, suppose that ties in bidding are broken randomly without bias.
- Suppose the seller announces a second-price sealed bid auction with reserve price of 1. Find a symmetric Bayesian Nash Equilibrium (BNE), $b^*(v)$, in which the bidders play weakly dominant strategies. How much expected revenue is generated in this BNE?
 - Suppose the seller announces a second-price sealed bid auction with reserve price of 1. Find an asymmetric BNE, $(b_1(v_1), b_2(v_2))$.
 - Suppose the seller announces a first-price sealed-bid auction with reserve price of 1. Find the symmetric BNE, $b^{**}(v)$ which generates the highest expected revenue. How much revenue is generated in the BNE?
 - Suppose the seller announces an all-pay auction with a reserve price of 1. Find a symmetric BNE, $b^{***}(v)$. How much expected revenue is generated in this BNE?
8. Sarah and Claire have identical utility functions $U = 5 \ln(x) + y$: Sarah starts out with 16 units of x and 5 units of y and Claire starts out with 4 units of x and 15 units of y .
- Draw the Edgeworth box, label the endowment point and sketch some "normal" indifference curves through the endowment point. Shade the region where trades are possible.
 - Find Sarah and Claire's Marshallian demands as a function of the price of x , price of y and their endowments.
 - Find the market clearing price and how much x and y the two consumer after trading.
 - Who is happier after trading?